

Total Error: What Classifier Validation Establishes About Exposure Comparisons

Gaurav Sood*

Abstract

A classifier’s confusion matrix need not validate comparisons constructed from its predictions. When domain labels are reused across browsing histories, the estimand determines a signed, weighted sum of domain mistakes. Its bias and variance depend on the full joint error distribution across domains. I characterize the sharp bounds on this error given exact target-population confusion counts and show when those counts identify a comparison. The decisive restriction is constant influence within each partially erroneous stratum. A finite-population audit gives simultaneous bounds for linear comparisons, but projecting count information can discard enough information that interval width remains positive even as the audit grows. Direct estimation of weighted errors avoids that loss. A reproducible browsing-data stress test illustrates the distinction, while an exact rare-error example shows why many contributing domains alone do not justify normal inference. The results clarify what validation must retain when predictions become research outcomes.

A researcher labels websites and combines those labels with browsing records to compare exposure across demographic groups. An accurate classifier seems to provide reassurance. Yet accuracy counts domains equally, whereas each comparison attaches its own signed weights to the same labels. A mistake on a domain visited disproportionately by one group can matter more than many correct labels elsewhere. Because users share labels and different domains can share classifier failures, measurement errors can be dependent even between users who visit different sites. Both bias and uncertainty depend on where errors occur.

This note asks what even an exactly known target-population confusion matrix establishes about such comparisons. The answer is a sharp identification interval obtained by sorting the comparison’s domain weights. The interval collapses precisely when weights are constant within every prediction stratum with an interior error count. This characterization separates a shortage of validation observations from information lost by summarizing those observations as counts. It also explains why a count-based audit can remain imprecise after the aggregate error fraction is accurately learned.

The problem intersects several established literatures. Quantification estimates population label prevalence from predictions and error rates (Vaz et al., 2019); prevalence uses equal domain weights, whereas exposure comparisons use heterogeneous, often signed weights. Shared domain errors have the same linear propagation structure that motivates attention to common shocks in shift-share inference (Adão et al., 2019). Multiaccuracy already expresses validation through residual moments for relevant functions (Kim et al., 2019). Design-based supervised learning supplies

*gsood07@gmail.com. The argument about browsing volume and costly errors on popular domains appears in the June 2020 manuscript coauthored with Suriyan Laohaprapanon (Laohaprapanon and Sood, 2020). The standalone note and original simulation were added in June 2023. This revision corrects the error-rate definitions and develops the validation bounds and inference results. Code and history: https://github.com/finite-sample/total_error.

unbiased corrections from probability samples of verified labels (Egami et al., 2023), and election and financial auditing already address bounded weighted errors using random audits (Stark, 2008; Shekhar et al., 2023). Neither covariance propagation nor the difference estimator is a new result here. The contribution is the explicit characterization of what confusion summaries retain for reused labels, its limits for auditing, and a reproducible assessment on exposure comparisons.

1 Start with the estimand

Suppose the research question concerns a difference in exposure between two groups, or a demographic coefficient in a regression of exposure. Define that quantity before assessing the classifier. Let $C = (c_{ij})$ contain visits by n users to k domains, and let $y_j \in \{0, 1\}$ be domain j 's true label. The true exposure vector is $T = Cy$. For declared user weights a , the estimand is

$$\theta = a^\top T = a^\top Cy = w^\top y, \quad w = C^\top a. \quad (1)$$

For a difference of group means, $a_i = 1/n_A$ in group A and $a_i = -1/n_B$ in group B . A single user's total sets that user's weight to one and all others to zero. For exposure shares, first replace C by $\text{diag}(V_1^{-1}, \dots, V_n^{-1})C$, where $V_i = \sum_j c_{ij} > 0$; the same mapping then gives the domain weights for the chosen share comparison. Replacing visits by duration changes the exposure unit but not the mapping.

Regression coefficients are also fixed linear functionals of the labels. For a full-rank demographic design X , write $A = (X^\top X)^{-1}X^\top$ and define the complete-label coefficient $\beta^* = ACy$. A coefficient contrast $r^\top \beta^*$ has user weights $a = A^\top r$ and domain weights $w = C^\top A^\top r$. These are descriptive coefficients for the specified browsing frame; causal or population interpretations require additional assumptions. Throughout, the browsing matrix, comparison weights, and regressors are fixed. Their uncertainty, if relevant to the research question, requires an additional sampling model. Domain labels also define the construct: they measure visits to domains with a given label, not necessarily consumption of that content on mixed-content sites.

Let $q \in \{0, 1\}^k$ contain predicted labels, $q_j = \hat{y}_j$, and let $e = q - y$. The predicted exposure vector is $\hat{T} = Cq$. The predicted comparison is $\hat{\theta} = w^\top q$, so its error is exactly

$$\Delta = \hat{\theta} - \theta = w^\top e. \quad (2)$$

Thus it is the estimand that determines which classification mistakes matter. A popular domain can dominate an individual's total but cancel from a group comparison if the groups use it equally. Conversely, a less popular domain visited disproportionately by one group can be influential for that comparison. An error rate that weights all domains equally need not describe either target.

For a fixed fitted classifier and fixed labels, Δ is fixed. To discuss bias and variance, specify a repeated-prediction process, such as repeated training samples, while fixing C, y, a and, where used, X, r . All moments below condition on this fixed design. Define $b = E[e]$ and the full joint covariance $\Sigma = \text{Cov}(e)$, without assuming independence across domains. Then

$$\begin{aligned} \text{Bias}(\hat{\theta}) &= w^\top b, \\ \text{Var}(\Delta) &= w^\top \Sigma w = \sum_j w_j^2 \Sigma_{jj} + 2 \sum_{j < \ell} w_j w_\ell \Sigma_{j\ell}, \end{aligned} \quad (3)$$

$$E[\Delta^2] = (w^\top b)^2 + w^\top \Sigma w. \quad (4)$$

Equations (1)–(4) connect the research quantity to its domain weights, its systematic error, and its uncertainty. Covariance between domain errors is part of the estimand’s error from the out-set. Dependence alone does not determine bias; systematically misclassifying domain types used disproportionately by one group can make $w^\top b$ nonzero.

User totals and regression vectors are corresponding joint quantities. Writing $d = \hat{T} - T$ and $\hat{\beta} = ACq$ gives

$$d = Ce, \tag{5}$$

$$E[d] = Cb, \quad \text{Cov}(d) = C\Sigma C^\top, \tag{6}$$

$$E[\hat{\beta} - \beta^*] = ACb, \quad \text{Cov}(\hat{\beta} - \beta^*) = AC\Sigma C^\top A^\top. \tag{7}$$

The same domain error is reused across all visits and users. These formulas quantify the specified prediction-error process; they do not include browsing-log error, user sampling, or uncertainty about the construct. They concern predicted exposure as an outcome. Noisy exposure as a regressor is a different errors-in-variables problem (Fong and Tyler, 2021).

2 Dependence follows the domain weights

There are two sources of dependence. A domain’s label is reused across users, and different domains are classified by the same fitted model. Shared training data, model parameters, and similar content or site templates can cause errors to move together under repeated fitting. Distinct domain names do not establish independent errors. Users need not visit the same domains to have correlated measurement errors: visiting different domains whose errors covary is enough. In equation (3), positively correlated mistakes increase variance when their domain weights have the same sign, and decrease it when the weights have opposite signs.

For a limiting example, suppose centered errors are shared within domain families: $e_j - b_j = u_{g(j)}$, with independent family shocks u_g of variance τ_g^2 . This is compatible with binary predictions when the classifier assigns one common random prediction to every domain in a family. Then

$$\text{Var}(\Delta) = \sum_g \tau_g^2 \left(\sum_{j:g(j)=g} w_j \right)^2. \tag{8}$$

A family of m domains with unit weights contributes $m^2\tau_g^2$, whereas assuming independent domain errors would give $m\tau_g^2$. Signed weights can instead cancel within a family. The relevant concentration is in aggregate family influence, not just individual domain influence. This example is a special covariance structure, not an assertion that the right families are known or that all within-family mistakes are perfectly correlated.

Independent domain errors are another special case: Σ is diagonal, with entries v_j . For user totals, equation (6) reduces to

$$\text{Var}(d_i) = \sum_j c_{ij}^2 v_j, \quad \text{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{9}$$

Let α_j be the false-positive probability on a truly negative domain and β_j the false-negative probability on a truly positive domain. For binary predictions, $v_j = \alpha_j(1 - \alpha_j)$ on negative domains and $v_j = \beta_j(1 - \beta_j)$ on positive domains. Repeated visits multiply one classification error; treating visits as independent would incorrectly replace squared visit weights by visit counts. Even this special case induces dependence across users who share uncertain domains.

An uncertainty procedure must follow the same structure as the estimand. Resampling users while holding predictions fixed does not propagate repeated- training uncertainty. Each replicate must preserve shared labels across users and the joint distribution of errors across domains. Independently perturbing each domain label imposes a diagonal Σ ; resampling the classifier fit and predicting all domains together can preserve dependence induced by that fitting process, provided the resampling scheme represents that process. A prediction-error covariance cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. Neither larger user samples nor user-level clustering alone guarantees that this error disappears, and standard errors do not repair bias.

2.1 Many domains do not ensure a normal approximation

Even with independent domain errors, diffuse variance shares alone do not justify Gaussian inference. Let all true labels be zero and let each of k domains be falsely labeled positive independently with probability $1/k$. With $w_j = 1$, total error B_k is binomial with mean one and variance $1 - 1/k$. The largest domain variance share is $1/k$, yet $B_k - 1 \Rightarrow \text{Poisson}(1) - 1$, not a normal distribution. At $k = 10,000$, an oracle interval using the exact bias and variance but a normal 1.96 cutoff has coverage 91.97%. This calculation uses exact binomial probabilities, not simulation.

For independent bounded binary errors, a sufficient Lindeberg condition is

$$\frac{\max_j |w_j|}{\sqrt{\sum_j w_j^2 v_j}} \rightarrow 0.$$

A small maximum variance share $\max_j w_j^2 v_j / \sum_j w_j^2 v_j$ is weaker when errors become rare. The example violates the former condition and satisfies the latter. A growing number of domains cannot substitute for a condition on the distribution of their errors. This is a prediction-error example, distinct from the fixed-label audit design.

3 What a confusion matrix identifies

A confusion matrix can be exactly correct for the target domains and still fail to determine a user comparison. This is an information problem before it is a sampling problem. Fix the predictions q and the domain weights w induced by the estimand in equation (1). We now ask what a validation summary establishes about the fixed target $\theta = w^\top y$, without specifying a stochastic model for how the classifier’s mistakes were generated.

Let $P = \{j : q_j = 1\}$ and $N = \{j : q_j = 0\}$. Suppose the entire target-population confusion matrix is known: f predictions in P are false positives and g predictions in N are false negatives. This grants more information than ordinary held-out validation, whose error rates require a transport assumption. For a finite collection B of weights, let $S_-(B, m)$ and $S_+(B, m)$ denote the sums of its m smallest and largest elements, with a zero sum when $m = 0$.

Proposition 1 (Sharp bounds from confusion counts). *Across binary label vectors compatible with q, f, g , the smallest and largest possible errors $\Delta = \hat{\theta} - \theta$ are*

$$\begin{aligned} L &= S_-(w_P, f) - S_+(w_N, g), \\ U &= S_+(w_P, f) - S_-(w_N, g). \end{aligned} \tag{10}$$

Consequently $[\hat{\theta} - U, \hat{\theta} - L]$ is the smallest interval containing every compatible value of θ . Both endpoints are attainable.

Proof. Only false positives and false negatives contribute: $\Delta = \sum_{j \in \text{FP}} w_j - \sum_{j \in \text{FN}} w_j$. Select f indices from P and g from N . The two selections have disjoint supports and no other restrictions. Selecting the smallest positive-stratum weights and largest negative-stratum weights minimizes the error; reversing these selections maximizes it. Assign the selected indices the corresponding true labels to attain each endpoint. $\square$

The compatible values form a finite set, so the interval is its convex hull; not every interior value need be attainable. A reported sign is established by these counts only if the bounding interval excludes zero. The calculation requires sorting, rather than a general integer-programming solver.

For example, take $q = (1, 1, 0, 0)$, $w = (1, -1, 1, -1)$, and $f = g = 1$. The same confusion matrix permits errors of -2 , 0 , and 2 . Prevalence, accuracy, FPR, and FNR are identical across those assignments. Adding correctly classified domains with zero contrast weight makes accuracy arbitrarily close to one without resolving the comparison. These are domains that can affect individual exposure but cancel from the chosen contrast.

Proposition 2 (When counts suffice). *In a stratum of M domains sharing the same predicted label, with exactly m errors, the error sum is point identified if and only if $m \in \{0, M\}$ or all its contrast weights are equal. Its identification width obeys*

$$S_+(w, m) - S_-(w, m) \leq \min(m, M - m) \left\{ \max_j w_j - \min_j w_j \right\}. \quad (11)$$

Widths add across strata. For several comparisons, counts determine every comparison if and only if their vector of domain weights is constant within each stratum with an interior error count.

Proof. Constant weights, no errors, and all errors each determine the sum. If $0 < m < M$ and two weights differ, choose an error set containing one of those indices but not the other. Exchanging them changes the sum, proving necessity. The difference between two sums of m terms is at most m times the range. Taking complements gives the corresponding bound with $M - m$ terms. The vector statement applies the exchange argument to each coordinate. $\square$

For a stratum mixing predicted labels, the corresponding criterion applies to the signed contributions $(2q_j - 1)w_j$, not to w_j alone. The result specifies what a finer validation table must preserve. Stratifying by predicted label alone is sufficient only when the contrast attaches equal weights within each partially erroneous stratum. Partitioning domains by similar contrast weights can reduce ambiguity; bins chosen for one comparison need not work for another. Error rates conditional on user demographics are not automatically available, because the same domain can enter many groups.

4 Auditing the target domains

A target-domain audit can convert these bounds into confidence intervals. The construction below uses established finite-population counting and audit ideas. Sorting worst-case error contributions and checking them by random audits has close precedents in election auditing (Stark, 2008). Weighted finite-population inference with prediction side information is also an established problem (Shekhar et al., 2023). The purpose here is to expose what confusion-count evidence preserves about reusable predictions, not to claim a new general audit principle or an optimal interval.

Partition the domains into strata $h = 1, \dots, H$ using information known before labels are audited. Draw a simple random sample without replacement of a fixed size n_h from the M_h domains in each

stratum. Let K_h be its unknown number of incorrect predictions and X_h the observed audit-error count. Conditional on the fixed labels and predictions,

$$X_h \sim \text{Hypergeometric}(M_h, K_h, n_h). \quad (12)$$

Choose nonnegative error budgets δ_h summing to at most δ . Retain each feasible integer K for which both $\Pr_K(X_h \leq x_h) \geq \delta_h/2$ and $\Pr_K(X_h \geq x_h) \geq \delta_h/2$. Write the smallest and largest retained integers as ℓ_h, u_h . A census stratum reveals $K_h = x_h$ exactly and needs no error budget; an unaudited stratum contributes no count restriction.

Let O be the audited domains and U_h the unaudited domains in stratum h . Set $c_j = (2q_j - 1)w_j$ and $d_O = \sum_{j \in O} w_j(q_j - y_j)$. Define

$$\begin{aligned} L_A &= d_O + \sum_h \min_{\ell_h - x_h \leq m \leq u_h - x_h} S_-(c_{U_h}, m), \\ U_A &= d_O + \sum_h \max_{\ell_h - x_h \leq m \leq u_h - x_h} S_+(c_{U_h}, m), \end{aligned} \quad (13)$$

where the optimization is over integers. Signed weights make this optimization necessary: an extreme need not occur at the largest admissible error count.

Proposition 3 (A simultaneous audit certificate). *Under the fixed stratified sampling design, the interval $[\hat{\theta} - U_A, \hat{\theta} - L_A]$ covers the fixed true contrast with probability at least $1 - \delta$. The same event guarantees coverage for every linear contrast of this one label vector. The endpoints are sharp over the binary configurations satisfying the count intervals and all audited labels.*

Proof. Inverting each pair of hypergeometric tails excludes the true K_h with probability at most δ_h . A union bound therefore includes all true counts with probability at least $1 - \delta$. On that event, the true errors in U_h number between $\ell_h - x_h$ and $u_h - x_h$. Sorting and choosing a count in that range gives the extrema in equation (13); the choices are independent restrictions across strata and hence jointly attainable. The true label vector lies in the same configuration set for every choice of w , which establishes simultaneous coverage. $\square$

The labels and predictions are fixed under this design: systematic or correlated classifier mistakes do not invalidate the guarantee. Coverage is over repeated audit samples, not conditional on the realized sampled identities. It requires neither independent classification errors nor a large-domain Gaussian approximation. The count confidence region can be conservative, and sharp projection of that region does not make the resulting confidence interval shortest. The guarantee concerns one outcome; combining separate audits of several outcomes requires an additional simultaneous-error allocation. Optional stopping, adaptive strata, and imperfect audit labels require further assumptions or other audit procedures.

The information discarded by count projection. A large audit need not make these intervals narrow. To see this, let all N predictions equal one, half the weights equal $1/N$ and half equal $-1/N$. Suppose a fraction $p < 1/2$ of labels in each half are wrong. The true contrast is zero. If $n \rightarrow \infty$ but $n/N \rightarrow 0$, an unstratified audit consistently learns the overall error fraction. Yet unaudited errors can still be placed almost entirely on either weight sign: the projected interval width converges to $2p$. A direct estimate of the weighted audit residual instead has standard error of order $n^{-1/2}$. Separating the two weight signs into strata removes this particular information loss. The distinction is between estimating a weighted residual and bounding it after retaining only counts.

For a declared contrast, an ordinary stratified difference estimator uses $z_j = w_j(y_j - q_j)$ directly:

$$\tilde{\theta} = \hat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} z_j, \quad \hat{V} = \sum_h \frac{M_h^2(1 - n_h/M_h)}{n_h} s_{z,h}^2, \quad (14)$$

where $s_{z,h}^2$ is the sample variance and a census stratum contributes zero variance. Under independent stratum draws, the estimate and variance estimator are design unbiased, with at least two audited domains in each noncensus stratum. Normal intervals need a separate approximation. A conservative finite-sample alternative applies Hoeffding’s without-replacement inequality (Hoeffding, 1963) to each bounded residual mean and sums the stratum radii. More efficient weighted audit procedures are available (Shekhar et al., 2023). The count certificate is useful for interpreting validation summaries and covering many comparisons together, not a general replacement for direct inference about a chosen comparison.

5 A browsing-data stress test

The empirical question is how much information confusion counts retain for comparisons supported by an actual browsing matrix. The fixed benchmark contains 1,132 pseudonymized users, 34,078 domains, 103,853 user–domain records, and 4,767,099 visits. It is the bundled `fewlab` extract of the June 2022 YouGov Pulse/RealityMine panel with Blacklight scanner outcomes. The replication package pins the source commit and file hashes. Its analytic universe contains domains with recorded scanner labels; exposure shares use visits within that universe as their denominator. These recorded outcomes are reference labels for the experiment, not error-free measures of all tracking activity. The exercise makes no causal or population-representative claim about demographic differences.

We retain the extract’s demographic design matrix: an intercept and indicators for gender, race, education, and source-coded age groups. Omitted categories are Male, White, HS or Below, and under 25. For each of seven scanner outcomes, we calculate all twelve non-intercept least-squares coefficients for both visit totals and visit shares. Each coefficient is a fixed weighted label sum. Two scanner columns are counts and are dichotomized at a positive value; the other five are binary. Complete labels, valid joins, positive visits, and full column rank are checked before estimation.

Predictions are imposed, rather than produced by a fitted classifier. Within each true class, we randomly flip the integer part of 1%, 5%, or 10% of the labels. We then perform two distinct exercises. First, holding those predictions fixed, Proposition 1 gives the true-coefficient range compatible with their exact confusion counts. Second, holding the reference labels fixed, we rearrange the same numbers of false positives and false negatives to minimize or maximize the predicted coefficient. The latter asks whether two classifiers with identical confusion matrices can report opposing comparisons. It is a worst-case sensitivity analysis, not an estimate of likely classifier bias. Extrema are computed separately for each coefficient; a single error arrangement need not reverse all comparisons at once.

At the smallest imposed error fraction, none of the 84 share-coefficient signs is identified by the full confusion matrix (Table 1); all permit a sign reversal by rearranging errors. The corresponding total coefficients show the same pattern, as do the larger error fractions. This establishes a limitation of confusion summaries for this frame. It does not establish that actual classifiers make such adverse errors. For example, the reference female coefficient for Facebook-pixel visit share is 0.828 percentage points. For the fixed random predictions, its count-compatible interval is $[-17.00, 16.72]$ percentage points. Holding the reference labels fixed instead, moving the same errors can produce predicted coefficients from -21.05 to 20.16 percentage points.

Table 1: What exact confusion counts establish at 1% imposed error in each true class

Scanner outcome	Sign identified	Reversal possible	Median width (pp)
Advertising trackers	0/12	12/12	47.8
Third-party cookies	0/12	12/12	52.5
Canvas fingerprinting	0/12	12/12	29.5
Session recording	0/12	12/12	31.6
Key logging	0/12	12/12	21.3
Facebook pixel	0/12	12/12	43.6
Google Analytics	0/12	12/12	21.6

Each row covers twelve demographic coefficients of visit shares in the fixed browsing frame. “Sign identified” counts bounding intervals excluding zero for fixed random predictions. “Reversal possible” counts coefficients for which rearranging errors with the same confusion matrix can reverse the reference coefficient’s sign. Median width summarizes identification intervals, measured in percentage points (pp); these are not sampling intervals. Error counts are rounded down within each true class. All outcomes, coefficients, scales, and error fractions are included in the machine-readable results.

5.1 What an audit recovers

The audit illustration uses the female coefficient for Facebook-pixel visit share. We fix one randomly constructed prediction vector with 5% errors in each true class. For budgets of 250, 1,000, and 2,000 domains, we compare stratified random sampling by predicted label with a design that first audits the most influential domains (rank of $|w_j|$) using half the budget. The remaining budget is allocated proportionally across the predicted-label tail strata and sampled randomly without replacement. All selection rules use known weights and predictions.

Each design is repeated 1,000 times with fixed labels and predictions. On every draw, the count projection, direct normal interval, and direct Hoeffding interval use exactly the same audited labels. All target 95% coverage; the first covers all linear contrasts of this outcome simultaneously, whereas the latter two target the declared coefficient. Hoeffding intervals are intersected with the logical range given the observed labels. This is a conservative finite-sample baseline; it is not a comparison against the most efficient available weighted-audit procedures.

Count projection covered the target in every replication but never excluded zero. Its intervals were much wider than the approximate normal intervals; the conservative direct Hoeffding intervals were wider still (Figure 1). Normal coverage in the smallest entirely random design was 93.1%, with Monte Carlo interval $[91.3, 94.6]\%$, below the nominal level. With 2,000 audited domains and half assigned to the certainty head, normal coverage was 95.6% and mean width was 1.40 percentage points. These results support retaining the locations and influences of errors, rather than treating their counts as a sufficient validation product. They neither establish normal validity for other outcomes nor an optimal allocation rule. Appendix C reports every design and method.

5.2 Finer validation summaries

Proposition 2 suggests retaining more information about influence within a validation table. In a follow-up after examining the initial audit results, we refine predicted-label strata by splitting the ranks of signed weights into one, five, or twenty-five bins. The resulting partitions have two, ten, or fifty strata. They use the same fixed prediction vector and coefficient as the audit illustration. For each partition, Table 2 separates the identification width if every stratum’s error count were known exactly from the mean confidence-interval width when those counts must be learned using 1,000 audited domains. The latter allocates samples proportionally across strata, with at least two observations per noncensus stratum, and uses 1,000 audit repetitions.

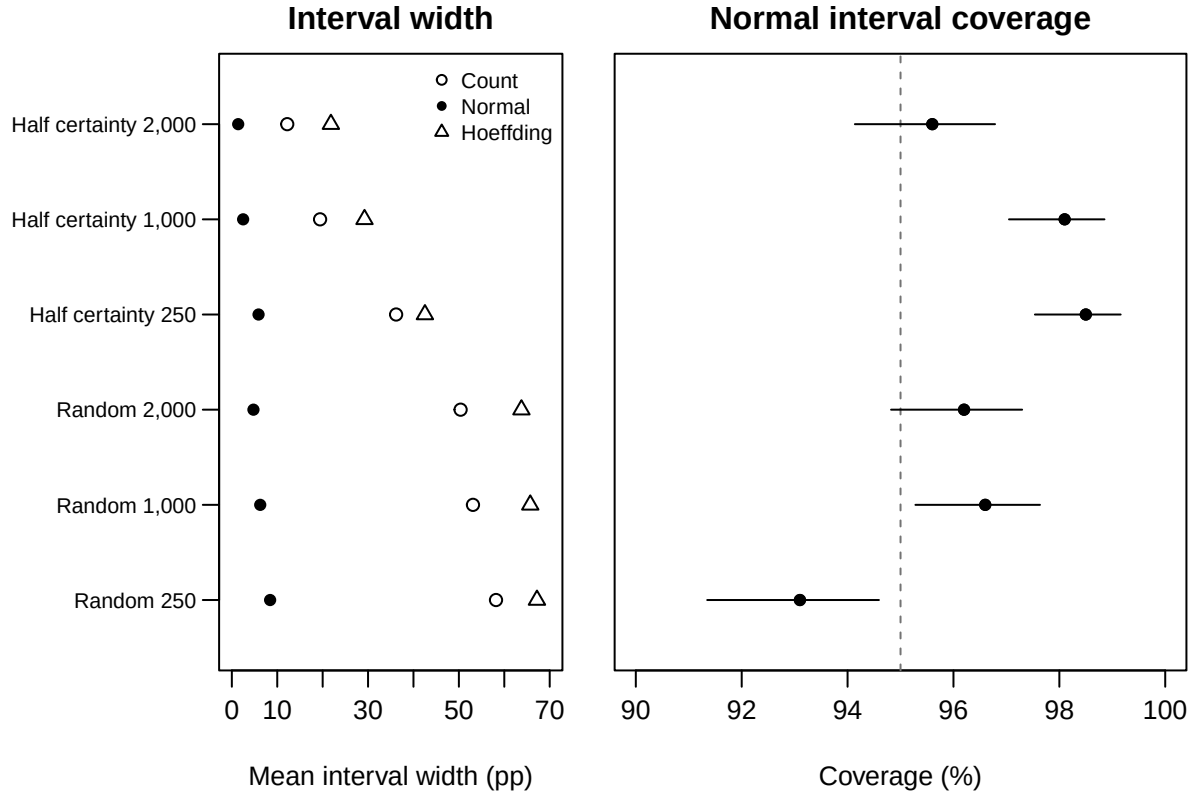


Figure 1: Width and coverage for the Facebook-pixel female share coefficient. Rows compare audit budgets and designs in the fixed browsing frame. Left: mean interval width in percentage points over 1,000 audits. Right: normal interval coverage, with exact 95% binomial Monte Carlo intervals and a dashed 95% reference. Count and Hoeffding intervals covered in every simulated draw. “Half certainty” audits the largest absolute influences with half the budget. Count intervals cover all contrasts of this outcome simultaneously; direct intervals target the declared coefficient. The randomness is domain auditing; users and predictions are held fixed.

Table 2: Refining validation reduces ambiguity but divides the audit budget

Strata	Exact-count width (pp)	Audit width (pp)	Coverage (%)
2	49.7	53.2	100.0
10	34.7	47.1	100.0
50	21.8	48.2	100.0

Facebook-pixel female share coefficient, fixed reference labels and imposed predictions. Rank bins of signed influence cross predicted labels. Exact-count width conditions on the population count in every stratum; audit width averages 95% count-projection intervals over 1,000 domain samples of size 1,000, with Bonferroni allocation across strata. Widths are percentage points. Coverage is empirical; the exact binomial Monte Carlo intervals accompany the results. No design certified the coefficient’s sign.

Finer summaries reduce identification ambiguity, as the proposition predicts. But the narrowest audit intervals among these partitions use ten strata, not fifty. Learning more stratum counts from a fixed audit budget and protecting their joint coverage offsets part of the gain. This comparison demonstrates both a constructive use of the characterization and its practical constraint: validation strata should preserve relevant weight differences without making each count too imprecise. It does not select an optimal partition.

6 Implications for correction and validation

Under the common-rate model in Appendix A, with known α, β and $\alpha + \beta \neq 1$, equation (A2) gives an unbiased correction:

$$\tilde{T}_i = \frac{\hat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \quad \mathbb{E}[\tilde{T}_i \mid C, y] = T_i. \quad (15)$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user’s weights.

Using probabilities q_j instead of hard labels yields $\sum_j c_{ij} q_j$. Under a conditional-label model with information $\mathcal{I}$ that includes browsing weights, $q_j = \Pr(y_j = 1 \mid \mathcal{I})$ makes that sum $\mathbb{E}[T_i \mid \mathcal{I}]$. Marginal calibration across domains alone does not supply this conditional property for every user or group. This conditional-label interpretation is distinct from the repeated-prediction interpretation used above; an uncertainty calculation must choose and state its target.

Validation should match the research quantity. For individual totals, large c_{ij} magnify label errors. For a contrast, the relevant quantities are $w^\top b$ and $w^\top \Sigma w$. Only under independent domain errors does variance reduce to separate contributions $w_j^2 v_j$; otherwise validation must also address which influential errors move together. These quantities can guide a labeling audit more directly than domain frequency alone. Verified labels for influential domains eliminate their classification error if those labels are correct. Usage-weighted training is another option, but its effect on exposure error requires validation; concentrated weights do not by themselves establish that model performance will worsen.

Changing classification thresholds is a sensitivity analysis, not a set of coverage-guaranteed bounds on true exposure. Winsorizing visits reduces the influence of large weights but changes the exposure quantity being estimated. The most useful validation reports errors in totals and in the intended comparisons, together with uncertainty that preserves the common domain labels.

References

- Rodrigo Adão, Michal Kolesár, and Eduardo Morales. Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, 134(4):1949–2010, 2019. doi: 10.1093/qje/qjz025.
- Naoki Egami, Musashi Hinck, Brandon Stewart, and Hanying Wei. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 68589–68601. Curran Associates, Inc., 2023. doi: 10.52202/075280-3000. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/d862f7f5445255090de13b825b880d59-Abstract-Conference.html.
- Christian Fong and Matthew Tyler. Machine learning predictions as regression covariates. *Political Analysis*, 29(4):467–484, 2021. doi: 10.1017/pan.2020.38.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- Michael P. Kim, Amirata Ghorbani, and James Zou. Multiaccuracy: Black-box post-processing for fairness in classification. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, pages 247–254, 2019. doi: 10.1145/3306618.3314287.
- Suriyan Laohaprapanon and Gaurav Sood. Domain knowledge: Predicting the kind of content hosted by a domain, 2020. URL https://github.com/themains/domain_knowledge/blob/850644c06d19a3d787a444443472367dd1bd3046/ms/domain_knowledge.tex. Manuscript version committed June 14. Section: Browsing Data: Concerns and Solutions.
- Shubhanshu Shekhar, Ziyu Xu, Zachary Lipton, Pierre Liang, and Aaditya Ramdas. Risk-limiting financial audits via weighted sampling without replacement. In *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, volume 216 of *Proceedings of Machine Learning Research*, pages 1932–1941, 2023. URL <https://proceedings.mlr.press/v216/shekhar23a.html>.
- Gaurav Sood. Estimand-driven sampling of reusable labels, 2026. URL <https://github.com/finite-sample/fewlab>. Unpublished working manuscript and companion software, version consulted October 4, 2026.
- Philip B. Stark. Conservative statistical post-election audits. *The Annals of Applied Statistics*, 2(2):550–581, 2008. doi: 10.1214/08-AOAS161.
- Afonso Fernandes Vaz, Rafael Izbicki, and Rafael Bassi Stern. Quantification under prior probability shift: the ratio estimator and its extensions. *Journal of Machine Learning Research*, 20(79):1–33, 2019. URL <https://www.jmlr.org/papers/v20/18-456.html>.

Appendix

A Bias and the denominators of error rates

Let α_j be the probability of predicting one for a truly negative domain and β_j the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1 - y_j) - \beta_j y_j, \quad \mathbb{E}[d_i | C, y] = \sum_j c_{ij} \{\alpha_j(1 - y_j) - \beta_j y_j\}. \quad (\text{A1})$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$\mathbb{E}[d_i | C, y] = \alpha(V_i - T_i) - \beta T_i = V_i \{\alpha - (\alpha + \beta)\pi_i\}, \quad \pi_i = T_i/V_i, \quad (\text{A2})$$

where the share π_i is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha + \beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does the number of distinct visited domains: repeated visits matter through c_{ij} .

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (A1), with rates conditional on those features, is the relevant expression. The bias in a share is $\mathbb{E}[d_i | C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

B A connection to label sampling

The same mapping supplies a remedy when a researcher can obtain verified labels for a probability sample of domains. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). Its regression decomposition is $H = AC$, with column h_j giving the influence of domain j on the coefficient. For exposure shares, replace C by its row-normalized version. For the scalar contrast in equation (2), the corresponding influence is w_j .

Freeze predictions q_j before drawing a label sample. Let R_j indicate selection, with known inclusion probability $s_j > 0$ for every domain. Conditional on the fixed population and predictions, the difference estimator is

$$\tilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \quad (\text{B1})$$

Since $\mathbb{E}_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier's errors. Verified labels must measure y_j correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli domain sampling,

$$\text{Var}_R(\tilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \quad (\text{B2})$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j)w_j^2(y_j - q_j)^2/s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with y and q fixed; it is distinct from the prediction-error process in equation (6). Both calculations preserve the reuse of a domain label across people.

C Full audit comparison

Table C1: All audit designs and methods for the focal coefficient

Design	Budget	Interval	Coverage (%)	Width (pp)	Sign (%)
Random	250	Count projection	100.0	58.2	0.0
Random	250	Direct normal	93.1	8.4	39.6
Random	250	Direct Hoeffding	100.0	67.2	0.0
Random	1,000	Count projection	100.0	53.1	0.0
Random	1,000	Direct normal	96.6	6.3	32.9
Random	1,000	Direct Hoeffding	100.0	65.7	0.0
Random	2,000	Count projection	100.0	50.4	0.0
Random	2,000	Direct normal	96.2	4.8	34.4
Random	2,000	Direct Hoeffding	100.0	63.8	0.0
Half certainty	250	Count projection	100.0	36.2	0.0
Half certainty	250	Direct normal	98.5	5.9	40.6
Half certainty	250	Direct Hoeffding	100.0	42.5	0.0
Half certainty	1,000	Count projection	100.0	19.4	0.0
Half certainty	1,000	Direct normal	98.1	2.5	41.9
Half certainty	1,000	Direct Hoeffding	100.0	29.2	0.0
Half certainty	2,000	Count projection	100.0	12.2	0.0
Half certainty	2,000	Direct normal	95.6	1.4	65.0
Half certainty	2,000	Direct Hoeffding	100.0	21.8	0.0

Each row summarizes 1,000 domain audits. Coverage and sign certification are percentages; width is the mean in percentage points. Sign certification means that the interval excludes zero, irrespective of whether it covers the target. All methods within a design use common draws. Exact binomial Monte Carlo intervals, standard errors, and replicate results accompany the package.

D The original simulation of accumulating bias

The original simulation draws 1,000 respondents, each with an independent observation count uniformly distributed over the integers from 5 to 1,000. Within respondents, independent bivariate normal pairs have zero means, unit variances, and correlation $\rho = 0.9$. The first component is truth and the second a proxy. The original code adds 0.1 times the respondent’s sample standard deviation of truth to every proxy observation. Table D1 retains those random draws and compares that specification with no added bias and an exactly constant addition of $b = 0.1$.

In the original specification, the observation-level correlation is 0.900 and the total correlation is 0.538. Mean total error is 51.239. Adding an exactly constant bias leaves the correlations of observations and respondent means unchanged. The SD-scaled addition only approximates a constant, so that invariance is not exact for the original specification.

Table D1: Correlations with truth and signed error in respondent totals

Proxy	Correlation			Mean total
	Observation	Mean	Total	error
No added bias	0.900	0.903	0.893	0.139
Constant bias	0.900	0.903	0.537	51.236
Original SD-scaled bias	0.900	0.901	0.538	51.239

Each row uses the same 1,000 respondents and random draws (seed 1234). Error is proxy minus truth. Measurements are centered continuous variables, so totals and errors are in arbitrary units. These are descriptive summaries of one simulated dataset; the table does not estimate uncertainty in browsing data.

The constant-bias design has a simple population expression. Let N be the observation count, independent of the pairs (Y_r, Z_r) , and set $S = \sum_{r=1}^N Y_r$, $\hat{S} = \sum_{r=1}^N (Z_r + b)$. Because the pairs have zero means and unit variances,

$$\begin{aligned} \text{Var}(S) &= \text{E}[N], & \text{Var}(\hat{S}) &= \text{E}[N] + b^2 \text{Var}(N), & \text{Cov}(S, \hat{S}) &= \rho \text{E}[N], \\ \text{Corr}(S, \hat{S}) &= \frac{\rho}{\sqrt{1 + b^2 \text{Var}(N)/\text{E}[N]}}. \end{aligned} \tag{D1}$$

The population correlation is 0.553, compared with 0.537 in the constant-bias simulation. With a fixed observation count, bias shifts every total equally and does not change the correlation. For nonzero means or a relation between counts and measurements, equation (D1) need not hold. Accumulating bias does not universally reduce correlations.

This simulation isolates bias accumulation. Its normal draws can be negative, so they are not literal visit counts or durations. It also generates observations independently across respondents and does not simulate shared domain errors. The covariance results above follow from the domain formulation, and are checked separately by enumerating all prediction vectors in a small binary example.